

Q.1

- 1.What is DDL and DML commands. Write syntax and examples for DDL and DML commands.
- 2.Discuss with suitable example Extended E-R features Specialization, Generalization and aggregation,
- 3.Explain any five relational algebra operators with suitable example.
- 4.Compare File Processing System with Database Management system
- 5.Define with an example different type of Entities in ER diagram
- 6.Define Triggers. Write syntax and example of trigger. of tris
- 7.Explain type of functional dependency with suitable example.
- 8.Describe types of Entities with example.
- 9.Identify different users of database management system
- 10.Explain all types of integrity constraints with an examples?
- 11.Discuss Log based recovery with an example
- 12.Explain the concept of data independence Discuss the differences between
- 13.logical and physical data independence
- 14.List and briefly explain SQL aggregate functions with suitable examples
- 15.Explain the concept of First Normal Form (1NF). Give example for the same.
- 16.Discuss conflict serializability with suitable example.
- 17.What is data abstraction
- 18.Draw Extended E-R diagram for Hospital Management System.

Q.2

- 1.Explain different types of users for database system and explain responsibilities of DBA.
- 2.What is Normalization. Explain 1 NF,2 NF,3NF with suitable example.
- 3.Design an EER diagram for Hospital Management System. And map it into relational model. Assume Suitable data.

4. Discuss three layer schema architecture with suitable diagram. What is Data Independence? Explain types of data independence.

5. What is deadlock? Give deadlock prevention methods with suitable example

6. Describe the overall architecture of DBMS with suitable diagram

7. Explain wait-die and wound-wait methods with suitable example

8. Explain types of joins with suitable example.

9. Write SQL expression for the following queries

Consider Employee relation (Emp_no, Emp_name, Emp_add, ph_no, email_id, salary)

Q.3

1. In a Hospital Management System there are many departments and many patients. Doctors work in various departments and treat multiple patients. Patients undergo multiple tests, and each test is conducted by lab technicians under a specific department.

2. Consider the above scenario and draw an E-R diagram. (Represent proper cardinalities)

3. Explain types of joins with suitable example.

A Consider the following employee database.

Employee (empname, street, city, date_of_joining)

Works (empname, company_name, salary)

any Company (company_name, city)

Manages (empname, manager_name)

Write the SQL queries for each of the statements given below

a) Modify the database so that 'John' now lives in 'Mumbai'

b) Find all employees who joined in the month of

c) Give all employees of 'ABC Corporation' a 10% raise.

d) Find all employees in the database who live in the same cities as the companies for which they work

e) Find all employees who earn more than average salary of all employees of their company

4. Construct an ER diagram and convert it into a relational model for a company which has several employees working on different types of Projects. Several employees are working for one department, every department has a manager. Several employees are supervised by one employee. Employees have zero or more dependents

5. Explain the following Relational Algebra operations with suitable example.

1) Generalized Project

2) Select

3) Union

4) Rename

5) Natural Join

6) set difference

6. Draw an E-R diagram for library management system. Convert it into relational schema

Q.4

1. What is 2PL. Explain its types.

2. Write SQL queries for the given database

Book(book id, title, author, cost)

Store(store no, city, state, inventory_val)

Stock(store_no, book_id, quantity)

(i) Modify the cost of DBMS books by 10%

(ii) Find the total number of books in Mumbai stores

(iii) Find title of all books whose title contains the word ' '

(iv) Find title of the most expensive book

(v) Add a new record in Book (Assume values as per requirement)

3. Write SQL queries for given database.

employee (eno, ename, bdate, title, salary, dno)

project (pno, pname, budget, dno)

department (dno, dname, mgrno)

works (eno, pno, resp, hours)

- 1) Find the project number and name for projects with a budget greater than 100,000.
- 2) Find the employees (name only) in department 'DI' ordered by descending salary.
- 3) Find all works records where hours worked is less than 10 and the responsibility is 'Manager'.
- 4) Find name of employees that are ending with letter 's'.
- 5) Find total number of employee.
- 6) Write the query to increase salary of the employees by 10%.
- 7) Find Employee name with maximum salary.

4. Explain all types of integrity constraints with an example?

5. Explain concurrency control and explain time Stamp based protocol of concurrency control.

Q.5

1. Write the SQL expression for the following.

customer (cust_id, cname, caddress, ph_no, balance)

- i) Find the names of customers whose name starts with the letter 'A'.
- ii) Add one record with the values (105, 'Rakhi', 'Malad', 9826756045, 3400).
- iii) Remove the data of the customer whose name is 'Abhay' and phone number is 8890654312.
- iv) Find the total number of customers.

2. Define view. Write syntax of creation of view considering suitable example.

3. Explain conflict serializability with suitable example.

4. Describe ACID properties with examples and explain state transition diagram of transaction.

5. Write note on

- i) Log based protocol
- ii) Timestamp-based protocols

6. Give example of serial schedule and equivalent to serial schedule with respect to conflict serializability. Discuss conflict serializability with example

7. Explain Lock based (2PL) concurrency control method with example

8. Design an EER schema for a BANK database.

9. Each bank can have multiple branches, and each branch can have multiple accounts and loans. Bank keeps the track of different types of Accounts (Saving_account, Checking_account), Loans (Car_loans, Home_loans, ...), each account's Transaction (deposit, withdrawal, check...) and each loan's Payments; both of these include the amount, date and time.

10. State any assumptions you make about the additional requirement clearly.

Q.6

1. Write note on any two

- a) ACID properties of transaction with
- b) Transaction state diagram and explain TCL commands
- c) Types of users in DBMS

2. Conversion of Specialization to relational schema with suitable example

- a) Log based recovery
- b) Role of DBA
- c) Types of attributes
- d) 2PL concurrency control protocol
- e) Triggers

f Lossless decomposition

3.Explain concept of Serializability along with Conflict Serializability and View

Serializability

4.Explain in detail with example of conflict and view serializability.

5.Explain following Integrity constraints:

a) Key Constraints.

b) Domain Constraints (Null & Default Constraints

c) Referential Constraints

d) Check Constraints.

6.Write short note on Log based recovery mechanism

7.Write a note on Time stamp ordering protocol.

List all functional dependencies satisfied by the relation.

X	Y	Z
X1	Y1	Z1
X1	Y2	Z1
X2	Y2	Z1
X2	Y2	Z1

g to search

T1	T2
read(A) A := A - 50	
	read(A) temp := A * 0.1 A := A - temp write(A) read(B)
write(A) read(B) B := B + 50 write(B) commit	
	B := B + temp write(B) commit

Draw the precedence graph for above schedule?

b Convert following E-R diagram to relational schema

